Weighted BDI Data Flow: Confidence, Priority, and Diagnostics

Author: CL.Investigate1 (Computational Linguist) **Date:** 2026-05-25 **Status:** Reference document

Overview

The debate engine uses two category-specific weights to control how taxonomy nodes influence agent behavior:

- **Belief Confidence** (0.0–1.0) — evidential strength of empirical claims
- **Desire Priority** (1–5) — normative importance of value commitments

These weights affect node sorting in prompts, inline labeling, agent framing instructions, and post-debate evolution. This document traces the complete data flow.

Pipeline Summary

```
INITIALIZATION                    DEBATE EXECUTION                  POST-DEBATE
──────────────                    ────────────────                  ───────────

assignWeights.ts                  debateEngine.ts
confidenceEvolution.ts
  │                                 │                                 │
  ├─ beliefConfidence.ts            ├─ taxonomyContext.ts             ├─ 3-
condition gate
  │    base + evidence +            │    weightedScore()              │
(attribution >0.6
  │    debate + edge boosts         │    nodeWeightLabel()            │     +
undermine
  │    → confidence [0.10,0.95]     │    formatTaxonomyContext()      │     +
strength >0.5)
  │                                 │      ↓                          │
  ├─ doctrinalAnchoring.ts          │    Sort: rel × confidence (B)   ├─
computeUndermineDelta
  │    cosine > 0.55 →              │          rel × priority/5 (D)   │    ±0.05–
0.10
  │    floor at 0.60                │      ↓                          │
  │                                 │    Label: "(confidence: 0.82)"  ├─ drift cap
±0.30
  ├─ desirePriority.ts              │           "(priority: 4/5)"     │
  │    tree position → 2–4          │           "[Speculative, 0.35]" ├─
confidenceDedup.ts
  │    doctrinal boundary → 5       │      ↓                          │    topic
dedup (cos>0.80)
  │                                 │    Instruction:                 │    attack
dedup (cos>0.85)
```

```
    └─ Write to POV JSON              │    "Lead with well-supported"    │    cross-
model robustness
                                      │    "Lead with core values"       │
                                      │      ↓                           └─ Write
updated nodes
                                      │    Agent prompt assembled             +
history entries
                                      │      ↓
                                      │    Debate runs...
                                      │      ↓
                                      └─ calibrationLogger.ts
                                           FIRE metrics, confidence
                                           deferrals, bottleneck ID
```

Stage 1: Initialization

1.1 Belief Confidence (`beliefConfidence.ts`)

`computeBeliefConfidence(signals)` applies a multi-signal formula:

```
confidence = base + evidenceBoost + debateBoost + edgeBoost
           → clamped [0.10, 0.95]
```

Signal	Source	Range
Base	`epistemic_type` × `falsifiability` from graph attributes	0.40–0.80
Evidence boost	+0.05 per unique source doc (capped +0.15)	0–0.15
Debate boost	+0.03 per prior debate reference (capped +0.10)	0–0.10
Edge boost	+0.02 per support edge, -0.02 per attack (capped ±0.05)	-0.05–+0.05

Base scores by epistemic type:

Epistemic Type	High Falsifiability	Medium	Low
`empirical_claim`	0.80	0.70	0.60
`predictive`	0.40	0.40	0.40
`interpretive_lens`	0.50	0.50	0.50
`definitional`	0.50	0.50	0.50

`assignBeliefConfidences()` runs this for all Belief nodes in a POV, sets `node.confidence` and creates the first `confidence_history` entry.

1.2 Doctrinal Anchoring (`doctrinalAnchoring.ts`)

Prevents core POV beliefs from being scored as speculative merely due to low source coverage.

1. Embed each POV's 4 doctrinal boundary strings (all-MiniLM-L6-v2, 384-dim)
2. For each Belief: compute cosine similarity to all boundary vectors
3. If max similarity ≥ **0.55** → `doctrinally_anchored = true`
4. If anchored AND confidence < **0.60**:
    - Save original as `evidential_confidence`
    - Raise `confidence` to the floor (0.60)

1.3 Desire Priority (`desirePriority.ts`)

`assignDesirePriorities()` scores based on tree position and doctrinal status:

Condition	Priority	Label
Doctrinal boundary	**5**	Core
Root node (no parent)	4	High
Mid-tree (parent + children)	3	Medium
Leaf node (parent, no children)	2	Low

1.4 Orchestration (`assignWeights.ts`)

CLI entry point: `npx tsx lib/debate/assignWeights.ts [--dry-run]`

Runs all three computations per POV, writes updated nodes back to disk. Outputs summary stats: avg/min/max confidence, anchoring counts, priority distribution.

Stage 2: Storage

PovNode Fields (`taxonomyTypes.ts`)

```
// Beliefs only
confidence?: number;                    // [0.0, 1.0]
confidence_history?: WeightHistoryEntry[];
doctrinally_anchored?: boolean;
evidential_confidence?: number;         // Pre-floor value when anchored

// Desires only
priority?: number;                      // 1–5
priority_history?: WeightHistoryEntry[];
```

Weight History Entry

```
interface WeightHistoryEntry {
  date: string;               // ISO 8601
  value: number;              // New value
```

```
  delta: number;              // Change from prior
  reason: string;             // "Initial multi-signal assignment" or "Debate X:
undermined"
  attack_claim?: string;      // AN claim text that triggered change
  supersedes?: string;        // Debate ID of prior update (dedup)
  robustness?: number;        // Models confirming (≥2 = cross-model)
  model_confirmations?: string[];
}
```

Stage 3: Prompt Injection

Weighted Sorting (`taxonomyContext.ts:88-97`)

```
function weightedScore(node, relevance, category) {
  if (Beliefs)    → relevance × confidence
  if (Desires)    → relevance × (priority / 5)
  if (Intentions) → relevance alone
}
```

High-confidence Beliefs and high-priority Desires sort to the top of their BDI sections.

Inline Labels (`taxonomyContext.ts:100-111`)

Condition	Label
Belief, confidence ≥ 0.50	`(confidence: 0.82)`
Belief, confidence < 0.50	`[Speculative, confidence: 0.35]`
Belief, doctrinally anchored	`(confidence: 0.60, doctrinally anchored)`
Desire	`(priority: 4/5)`
Intention	*(no label)*

Framing Instructions (`taxonomyContext.ts:148-152`)

When weighted data is present, the formatter injects category-specific instructions:

- **Beliefs**: "*Ordered by evidential confidence. Lead with well-supported claims. When you cite a low-confidence Belief, acknowledge the uncertainty explicitly.*"
- **Desires**: "*Ordered by priority. Lead with non-negotiable values. Lower-priority desires may be traded off in argument.*"

Example Prompt Output

```
=== YOUR EMPIRICAL GROUNDING (what you take as true) ===
Ordered by evidential confidence. Lead with well-supported claims.
```

```
★ [acc-bel-042] (relevance: 0.71) (confidence: 0.82) Inherent Power-Seeking: ...
★ [acc-bel-015] (relevance: 0.68) (confidence: 0.75, doctrinally anchored) Market
Efficiency: ...
  [acc-bel-088] (relevance: 0.52) [Speculative, confidence: 0.35] Rapid Capability
Emergence: ...

=== YOUR NORMATIVE COMMITMENTS (what you argue should happen) ===
Ordered by priority. Lead with non-negotiable values.

★ [acc-des-001] (relevance: 0.80) (priority: 5/5) Innovation Freedom: ...
★ [acc-des-013] (relevance: 0.65) (priority: 4/5) Long-Term AI Benefit: ...
  [acc-des-027] (relevance: 0.48) (priority: 2/5) Cost Reduction: ...
```

Calibration Audit (`prompts.ts:2121-2172`)

When belief confidences are available, the extraction/evaluation prompt includes a calibration block that asks the LLM to verify whether the draft's rhetoric matches evidential strength — citing a low-confidence Belief as "established fact" is flagged as a weakness.

Stage 4: Mid-Debate Evolution

Confidence Updates (`confidenceEvolution.ts`)

After a debate completes, the engine evaluates whether Belief confidences should change.

Three-condition gate (all must pass to reduce confidence):

1. `attribution_confidence > 0.60` — the AN claim clearly instantiates the Belief
2. `attack_type === 'undermine'` — the attack targets evidential foundations (not just conclusions)
3. `attack_strength > 0.5` — the attack is strong enough to matter

Update magnitude (`computeUndermineDelta`):

Claim Outcome	Strength	Delta
Claim survived attacks	> 0.7	**+0.05**
Claim in contested middle	0.3–0.7	**0**
Claim fell below viability	< 0.3	**-0.05 to -0.10**

Drift cap: Total change from initial assignment capped at ±0.30 (`MAX_DRIFT`). Prevents a single debate from radically altering long-held positions.

Additional signals:

- Cross-POV validation (opposing camp cites Belief as evidence): **+0.10**
- Document contradiction discovered: **-0.15**

Priority Updates

- **Concession during debate**: reduce priority by 1
- **Identified as crux of disagreement**: increase priority by 1

Cross-Debate Deduplication (`confidenceDedup.ts`)

Prevents multi-model debate runs from double-penalizing the same Belief.

Check	Similarity Threshold	Behavior
Same topic	cosine > 0.80	Take max magnitude; discard weaker
Same attack vector, same model	cosine > 0.85	Replace if stronger; discard if weaker
Same attack vector, different model	cosine > 0.85	**Robustness signal** — record both models confirming

Cross-model robustness (≥2 models confirming the same attack) is recorded in the history entry as `robustness: N` and `model_confirmations: [...]`.

History Pruning

Keep last 30 entries OR entries ≤12 months old. Track pruned net delta for audit trail.

Stage 5: Diagnostics Surfacing

DiagnosticsWindow — Confidence Impact Panel (`DiagnosticsWindow.tsx:3119-3175`)

After a debate loads, scans all POV taxonomy nodes for `confidence_history` and `priority_history` entries matching the debate ID. Renders a panel showing:

```
Confidence Impact (5)
  acc-bel-042    Inherent Power-Seeking       0.82    +0.05    Survived attacks
  saf-des-015    Mitigating Displacement       3       -1       Concession
  skp-bel-068    Long-Horizon Uncertainty     0.45    -0.08    Undermined
```

- Delta colored green (+), red (-), or gray (0)
- Robustness badge: "2× confirmed" for cross-model agreement
- Attack claim snippet shown for undermine-driven changes

Node Detail View (`GraphAttributesPanel.tsx`)

Belief nodes:

- Confidence slider (0.00–1.00), color-coded: green ≥0.70, blue 0.40–0.69, orange <0.40
- Doctrinal anchor badge with tooltip explaining the floor mechanism
- Evidential confidence display when floor was applied: "*Evidential: 0.35 (floor applied: 0.60)*"
- History summary: "*3 update(s) — latest: Debate d-2024-11-18: survived*"

Desire nodes:

- Priority dropdown (1–5)
- P5 flagged: "Core — doctrinal boundary"
- History summary with latest reason

Calibration Log (`calibrationLogger.ts`)

Each debate session appends to `calibration-log.json`:

- `fire_confidence_threshold` — current FIRE acceptance threshold
- `confidence_deferrals` — times extraction confidence was deferred
- `confidence_bottleneck` — which signal bottlenecked ('extraction' | 'stability' | 'none')
- `fire_survived_rate` — fraction of borderline (0.5–0.7) claims that survived debate
- Used by the calibration optimizer for real-time threshold auto-tuning

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File Reference

File	Key Functions	Stage
`beliefConfidence.ts`	`computeBeliefConfidence()`, `assignBeliefConfidences()`	Init
`desirePriority.ts`	`computeTreePriority()`, `assignDesirePriorities()`	Init
`doctrinalAnchoring.ts`	`computeDoctrinalAnchoring()`	Init
`assignWeights.ts`	CLI orchestrator	Init
`taxonomyTypes.ts:59-86`	`PovNode` fields, `WeightHistoryEntry`	Storage
`taxonomyContext.ts:88-190`	`weightedScore()`, `nodeWeightLabel()`, `formatTaxonomyContext()`	Prompt
`prompts.ts:2121-2172`	Confidence calibration audit block	Prompt
`confidenceEvolution.ts`	`evaluateGate()`, `computeUndermineDelta()`, `pruneHistory()`	Evolution
`confidenceDedup.ts`	`evaluateDedup()`, cross-model robustness	Evolution
`DiagnosticsWindow.tsx:3119-3175`	Confidence impact panel	Diagnostics
`GraphAttributesPanel.tsx`	Sliders, badges, history summary	Diagnostics
`calibrationLogger.ts`	FIRE metrics, confidence deferrals	Diagnostics

Design Principles

1. **Weights are signals, not gates.** Low confidence doesn't block a Belief from injection — it changes its sort position and label. The agent sees it and can choose to cite it with appropriate hedging.

2. **Doctrinal anchoring prevents epistemic bootstrapping.** A core POV belief shouldn't be scored as "speculative" just because it has fewer source docs than an empirical claim. The floor ensures doctrinal positions maintain minimum presence.

3. **Evolution is conservative.** The 3-condition gate + drift cap ensures that confidence changes are evidence-driven and bounded. A single bad debate can't destroy a well-supported Belief.

4. **Cross-model robustness adds signal, not penalty.** When two different models independently confirm the same attack on a Belief, that's recorded as a robustness signal — stronger evidence than a single model's judgment.

5. **Priority is structural, not evidential.** Desire priority comes from tree position and doctrinal status, not from how many sources mention it. This reflects the normative nature of Desires — they're commitments, not hypotheses.